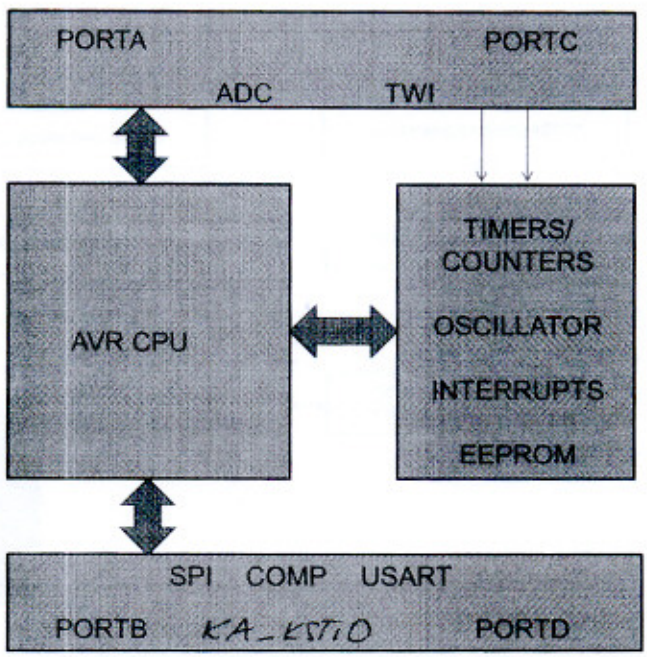
STRUKTURY WBUDOWANE

2

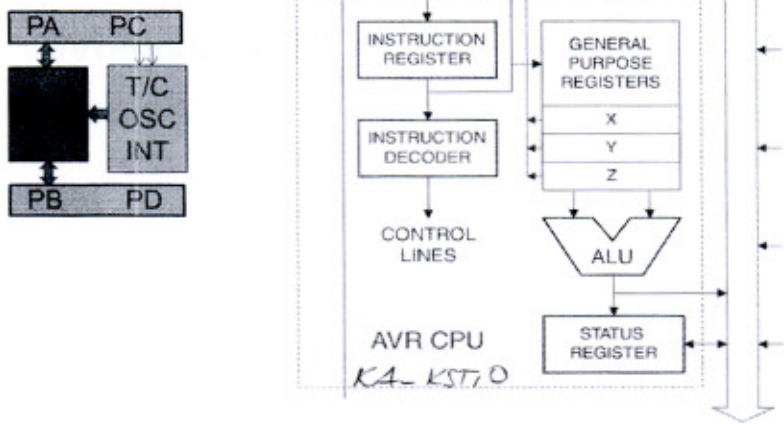
Oprac. K. Arnold

KSTiO

ATMEGA16A

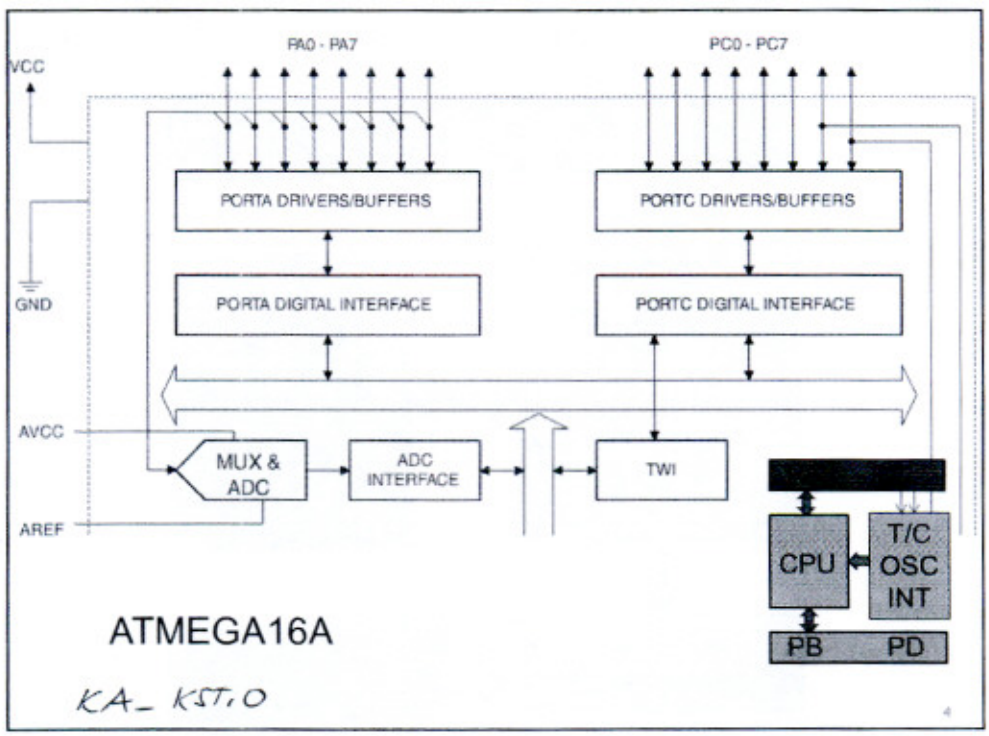


ATMEGA16A



KA_ KST, O

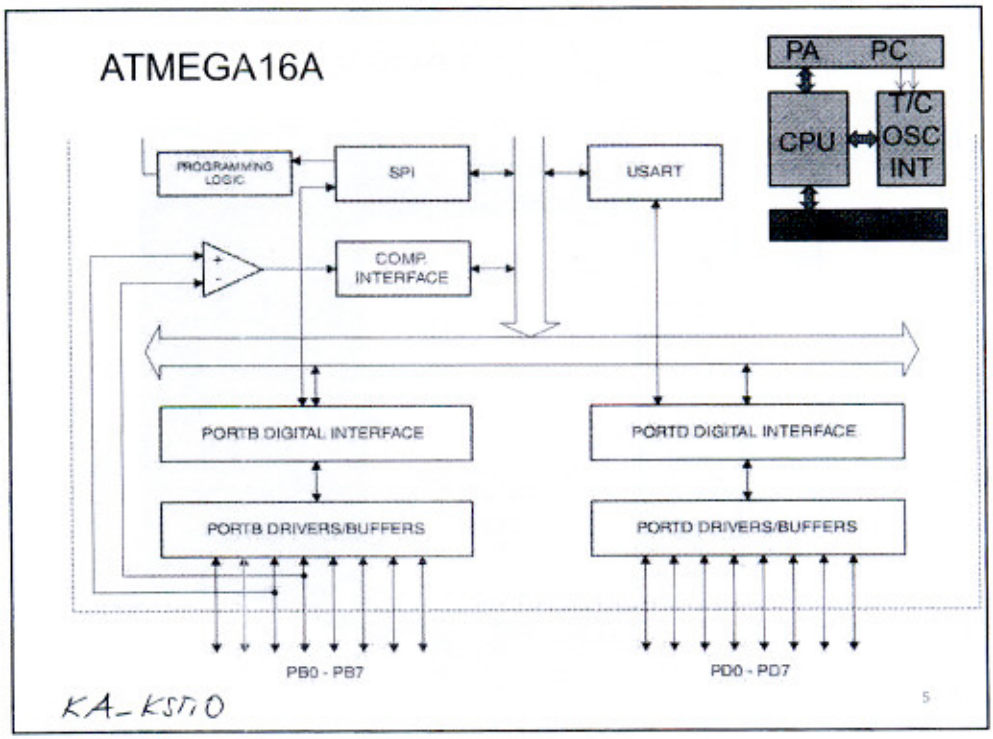
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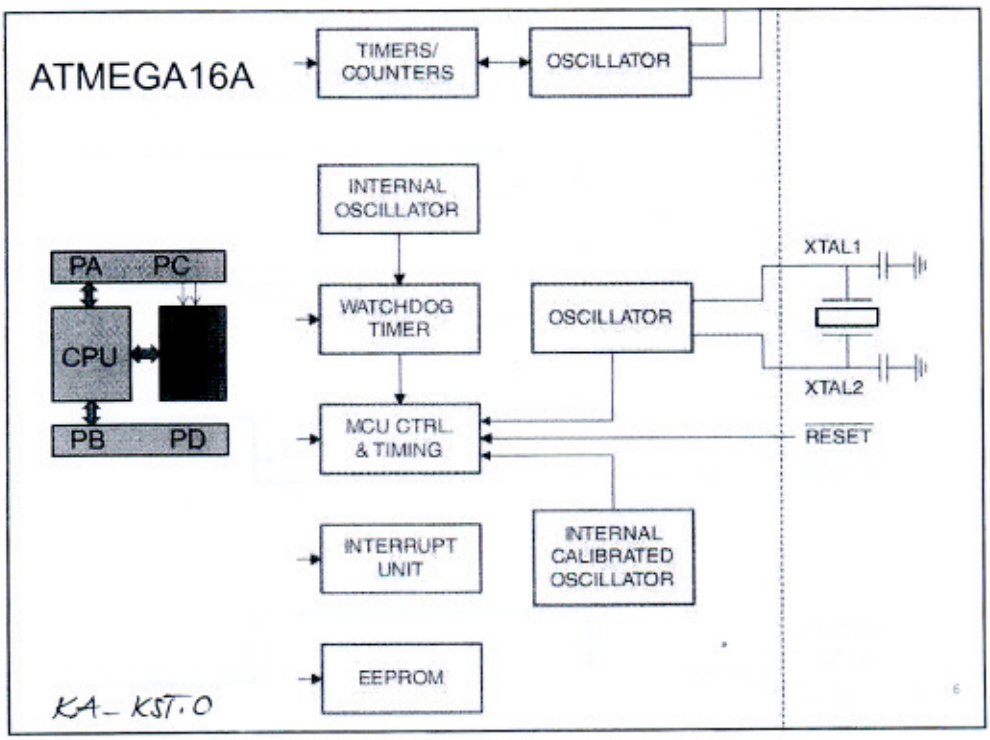


ATMEGA16A

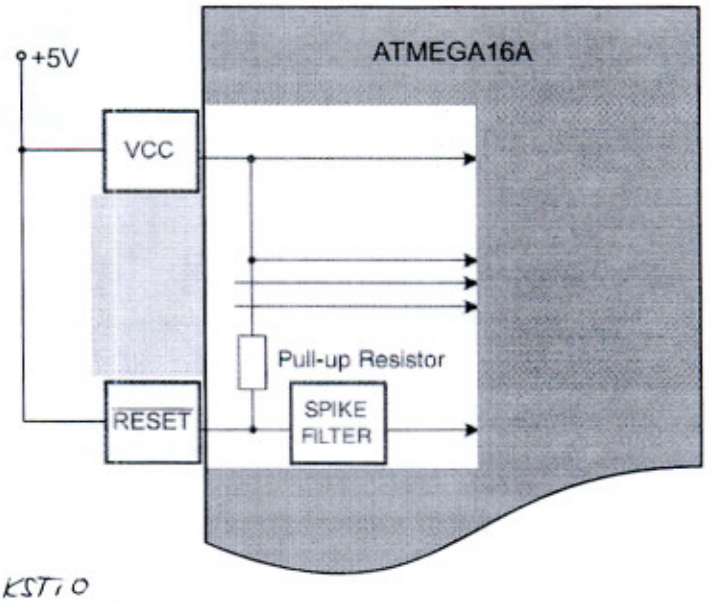
KA_ KST, O

4


ATMEGA16A
PA
PC
CPU
T/C
OSC
INT
PROGRAMMING LOGIC
SPI
USART
COMP. INTERFACE
PORTB DIGITAL INTERFACE
PORTD DIGITAL INTERFACE
PORTB DRIVERS/BUFFERS
PORTD DRIVERS/BUFFERS
PB0 - PB7
PD0 - PD7
KA_KST.O
5

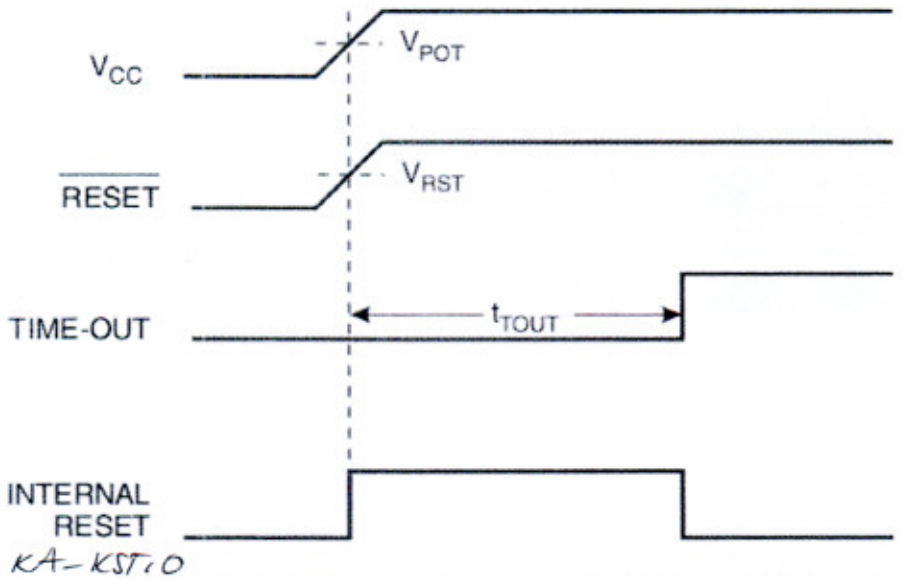

ATMEGA16A
TIMERS/ COUNTERS
OSCILLATOR
INTERNAL OSCILLATOR
PA
PC
CPU
PB
PD
WATCHDOG TIMER
OSCILLATOR
XTAL1
XTAL2
MCU CTRL. & TIMING
RESET
INTERRUPT UNIT
INTERNAL CALIBRATED OSCILLATOR
EEPROM
KA_KST.O
6

POR RESET AVR MCU



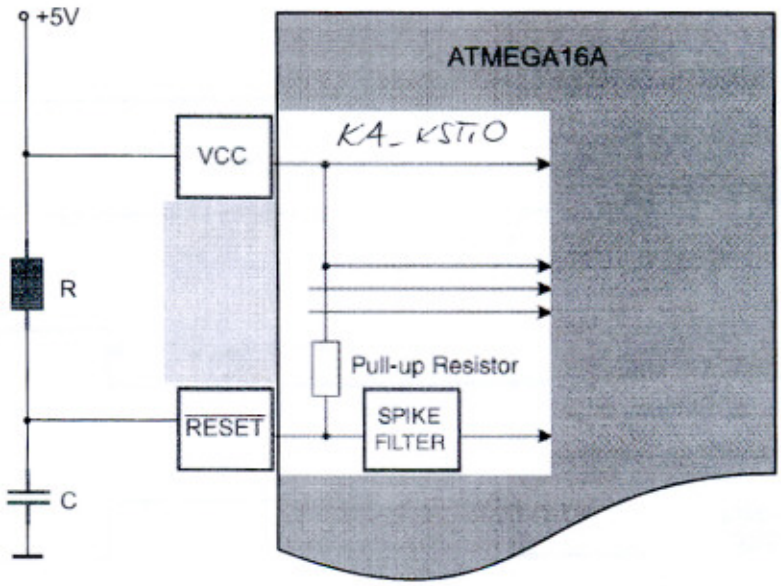
KA-KST:0

MCU Start-up, RESET Tied to VCC



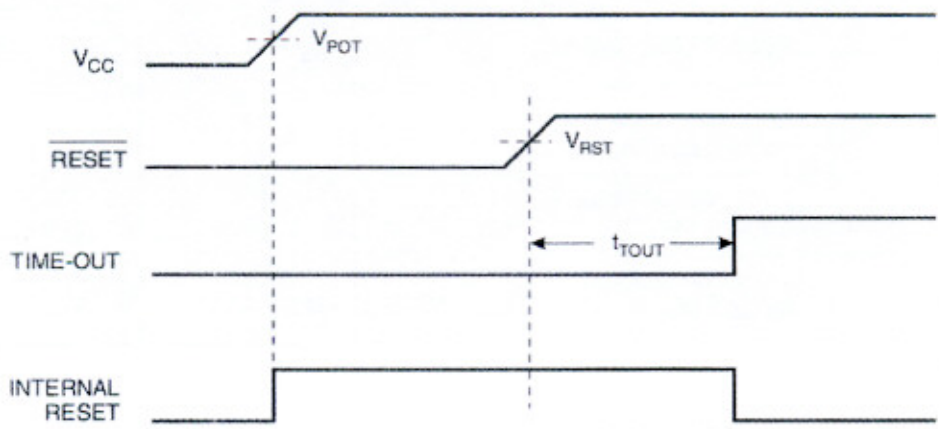
KA-KST:0

EXTERNAL RESET AVR MCU



9

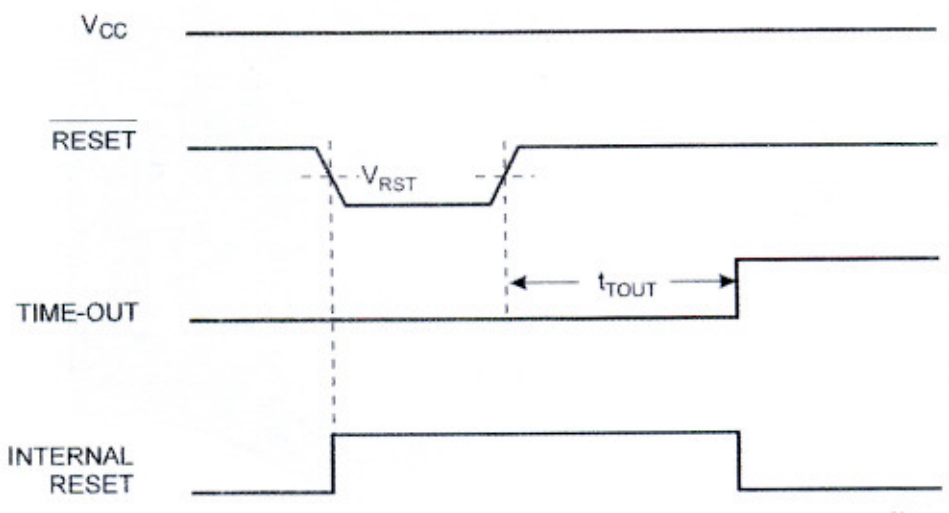
MCU Start-up, RESET Extended Externally



KA_KST.0

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External Reset During Operation



KA_KST10

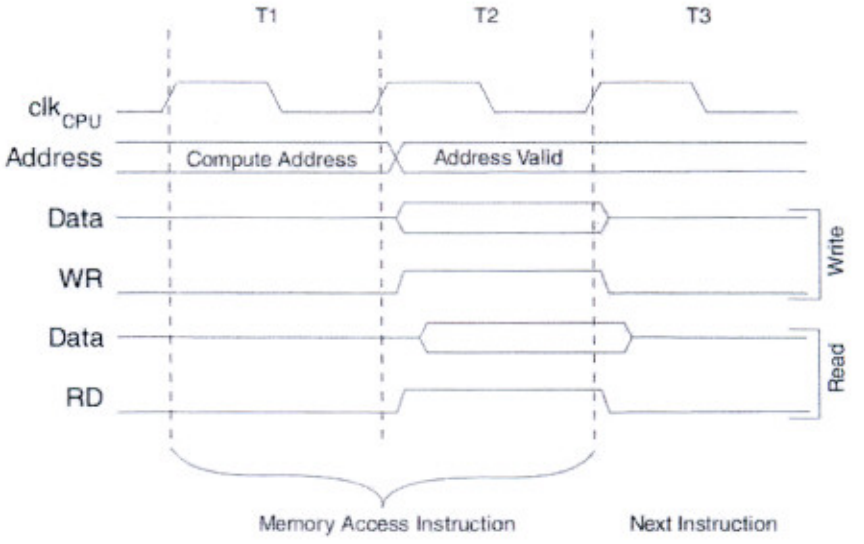
RESET Pin

Symbol	Parameter	Condition	Min	Typ	Max	Unit
R_{RST}	Reset Pull-up Resistor		30	60	85	kΩ
t_{RST}	Minimum pulse width on RESET Pin		1.5			μs
V_{IH2}	Input High Voltage RESET Pin	V_{CC}= 2.7- 5.5	0.9 V_{CC}		V_{CC}+0.5	V
V_{IL2}	Input Low Voltage RESET Pin	V_{CC}= 2.7- 5.5	- 0.5		0.2 V_{CC}	V

- RC = 100nF * 30kΩ = 3ms
- RC = 47nF * 30kΩ = 1.4ms

KA_KST10

On-chip Data SRAM Access Cycles



KA_ KST 1,0

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STOSOWANIE LDI, MOV, ROR, DEC

R0	1000 0110
R1	0000 0000
R2	0000 0010
R16	0000 1100
R17	0000 0001
R18	0000 0010

```
.org 0x00
     jmp MAIN

.org 0x2A
MAIN:
     ldi r16, 0x0C     ; ładuj dzielną
     ldi r17, 0x01
     ldi r18, 0x02     ; wykładnik dzielnika
     movw r0, r16      ; kopiuj rejestry
     mov r2, r18
DIV:
     clc               ; zeruj bit C w SREG
     ror r1            ; przesuń w prawo
     ror r0
     dec r2            ; zmniejsz licznik
                       ; przesunięć
     brne DIV          ; powtórz, jeśli Z=0
END_DIV:
```

KA_ KST 1,0

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